

- 1 (a) Describe how a decrease in the water potential of the blood leads to an increase in the concentration of antidiuretic hormone (ADH) in the blood.

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..... [3]

- (b) ADH is transported in the blood to the cells of the collecting duct of the kidney.

Fig. 1.1 is a diagram outlining the action of ADH on the cells of the collecting duct.

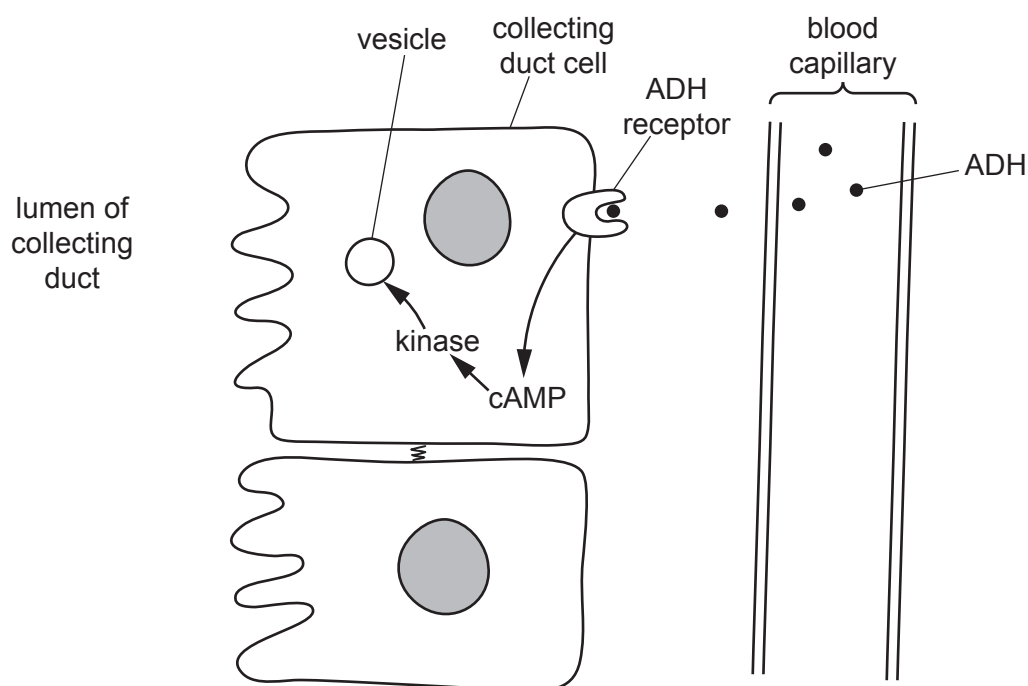


Fig. 1.1

- (i) Binding of ADH to the ADH receptor triggers reactions within the cell. One of the first reactions is the production of cyclic AMP (cAMP).

State the term used to describe molecules such as cAMP in cell signalling.

..... [1]

(ii) Describe the role of the vesicle when stimulated by kinase enzyme.

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..... [2]

(iii) On Fig. 1.1, draw arrows to show the direction of movement of water molecules after the cells of the collecting duct have responded to ADH. [1]

(c) Some people have a rare kidney disorder in which ADH is not able to bind to ADH receptors. Suggest the effects of this disorder on osmoregulation.

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..... [2]

[Total: 9]